SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN

Science, Mathematics and Technology

SSILP Math

Beginner

LAWS OF LOGARITHMS

EXPONENTIAL & LOG EQUATIONS

Learning Objectives

- Understand the fundamental laws of logarithms and their applications.
- Learn how to use the laws of logarithms to evaluate and simplify expressions.
- Gain proficiency in expanding and combining logarithmic expressions.
- Identify common errors to avoid when working with logarithms.
- Master the change of base formula for logarithms.
- Solve exponential equations by understanding the properties of exponential functions.
- Solve logarithmic equations and understand their properties.
- Tackle word problems involving exponential and logarithmic equations.

Session Outline

1 Laws of Logarithms

Laws of logarithms

LAWS OF LOGARITHMS

Let a be a positive number, with $a \neq 1$. Let A, B, and C be any real numbers with $A > 0$ and $B > 0$.

Law	Description
1. $\log_a(AB) = \log_a A + \log_a B$	The logarithm of a product of numbers is the sum of the logarithms of the numbers.
2. $\log_a\left(\dfrac{A}{B}\right) = \log_a A - \log_a B$	The logarithm of a quotient of numbers is the difference of the logarithms of the numbers.
3. $\log_a(A^C) = C\log_a A$	The logarithm of a power of a number is the exponent times the logarithm of the number.

Using the Laws of Logarithms to Evaluate Expressions

Evaluate each expression.

(a) $\log_4 2 + \log_4 32$

(b) $\log_2 80 - \log_2 5$

(c) $-\frac{1}{3} \log 8$

SOLUTION

(a) $\log_4 2 + \log_4 32 = \log_4(2 \cdot 32)$ Law 1

$= \log_4 64 = 3$ Because $64 = 4^3$

(b) $\log_2 80 - \log_2 5 = \log_2\left(\frac{80}{5}\right)$ Law 2

$= \log_2 16 = 4$ Because $16 = 2^4$

(c) $-\frac{1}{3} \log 8 = \log 8^{-1/3}$ Law 3

$= \log\left(\frac{1}{2}\right)$ Property of negative exponents

EXAMPLE

Expanding Logarithmic expressions

Use the Laws of Logarithms to expand each expression.

(a) $\log_2(6x)$ **(b)** $\log_5(x^3y^6)$ **(c)** $\ln\left(\dfrac{ab}{\sqrt[3]{c}}\right)$

SOLUTION

(a) $\log_2(6x) = \log_2 6 + \log_2 x$ Law 1

(b) $\log_5(x^3y^6) = \log_5 x^3 + \log_5 y^6$ Law 1

$= 3\log_5 x + 6\log_5 y$ Law 3

(c) $\ln\left(\dfrac{ab}{\sqrt[3]{c}}\right) = \ln(ab) - \ln\sqrt[3]{c}$ Law 2

$= \ln a + \ln b - \ln c^{1/3}$ Law 1

$= \ln a + \ln b - \frac{1}{3}\ln c$ Law 3

EXAMPLE

Combining Logarithmic expressions

Use the Laws of Logarithms to combine each expression into a single logarithm.

(a) $3 \log x + \frac{1}{2} \log(x + 1)$

(b) $3 \ln s + \frac{1}{2} \ln t - 4 \ln(t^2 + 1)$

SOLUTION

(a)
$$\begin{aligned} 3 \log x + \tfrac{1}{2} \log(x + 1) &= \log x^3 + \log(x + 1)^{1/2} && \text{Law 3} \\ &= \log(x^3(x + 1)^{1/2}) && \text{Law 1} \end{aligned}$$

(b)
$$\begin{aligned} 3 \ln s + \tfrac{1}{2} \ln t - 4 \ln(t^2 + 1) &= \ln s^3 + \ln t^{1/2} - \ln(t^2 + 1)^4 && \text{Law 3} \\ &= \ln(s^3 t^{1/2}) - \ln(t^2 + 1)^4 && \text{Law 1} \\ &= \ln\left(\frac{s^3\sqrt{t}}{(t^2 + 1)^4}\right) && \text{Law 2} \end{aligned}$$

EXAMPLE

Logarithms Common errors to avoid

Warning Although the Laws of Logarithms tell us how to compute the logarithm of a product or a quotient, *there is no corresponding rule for the logarithm of a sum or a difference*. For instance,

$$\log_a(x + y) \cancel{=} \log_a x + \log_a y$$

In fact, we know that the right side is equal to $\log_a(xy)$. Also, don't improperly simplify quotients or powers of logarithms. For instance,

$$\frac{\log 6}{\log 2} \cancel{=} \log\left(\frac{6}{2}\right) \qquad \text{and} \qquad (\log_2 x)^3 \cancel{=} 3 \log_2 x$$

Change of base formula

Suppose we are given $\log_a x$ and want to find $\log_b x$. Let

$$y = \log_b x$$

We write this in exponential form and take the logarithm, with base a, of each side.

$$\begin{aligned} b^y &= x && \text{Exponential form} \\ \log_a(b^y) &= \log_a x && \text{Take } \log_a \text{ of each side} \\ y \log_a b &= \log_a x && \text{Law 3} \\ y &= \frac{\log_a x}{\log_a b} && \text{Divide by } \log_a b \end{aligned}$$

CHANGE OF BASE FORMULA

$$\log_b x = \frac{\log_a x}{\log_a b}$$

In particular, if we put $x = a$, then $\log_a a = 1$, and this formula becomes

$$\log_b a = \frac{1}{\log_a b}$$

Use Laws of Logarithms to evaluate

(a) $\log_3 135 - \log_3 45$

(b) $-\frac{1}{3} \log_3 27$

(c) $\log_3 100 - \log_3 18 - \log_3 50$

ACTIVITY 1.1

1.1 Use Laws of Logarithms to evaluate

(a) $\log_3 135 - \log_3 45$

$\log_3\left(\frac{135}{45}\right)$

$\log_3 3 = \log_3 3^1$

$= 1$

(b) $-\frac{1}{3}\log_3 27$

$-\frac{1}{3}\log_3 3^3$

$\log_3 3^{3/-3} = \log_3 3^{-1}$

$= -1$

(c) $\log_3 100 - \log_3 18 - \log_3 50$

To keep it even we apply the Second Law of Logarithms to $\log_3 100$ - $\log_3 50$

$\log_3 \frac{100}{50}$ - $\log_3 18$

$\log_3 2$ - $\log_3 18$

Apply the Second Law of Logarithms again

$\log_3 \frac{2}{18}$

$\log_3 \frac{1}{9}$

$\log_3 \frac{1}{3^2}$

Move the denominator to the numerator with a negative exponent

$\log_3 3^{-2}$

$= -2$

SOLUTION

Expanding Logarithmic expressions

(a) $\log_3\left(x\sqrt{y}\right)$

(b) $\log \sqrt[3]{x^2 + 4}$

(c) $\ln \dfrac{3x^2}{(x+1)^{10}}$

ACTIVITY 1.2

1.2 Expanding Logarithmic expressions

(a) $\log_3(x\sqrt{y})$

Expand the expression:
$\log_3(x\sqrt{y})$

Rewrite the root for y
$\log_3(xy^{\frac{1}{2}})$

Apply the First Law of Logarithms
$\log_3(x \times y^{\frac{1}{2}}) = \log_3 x + \log_3 y^{\frac{1}{2}}$

Apply the Third Law of Logarithms for $\log_3 y^{\frac{1}{2}}$
$\log_3 y^{\frac{1}{2}} = \frac{1}{2}\log_3 y$
$\log_3 x + \log_3 y^{\frac{1}{2}} = \log_3 x + \frac{1}{2}\log_3 y$

(b) $\log \sqrt[3]{x^2 + 4}$

Expand the expression:
$\log \sqrt[3]{x^2 + 4}$

Rewrite the cube root
$\log(x^2 + 4)^{\frac{1}{3}}$

Apply the Third Law of Logarithms
$\log(x^2 + 4)^{\frac{1}{3}} = \frac{1}{3}\log(x^2 + 4)$

No further expansion can take place

$\frac{1}{3}\log(x^2 + 4)$

SOLUTION

1.2 Expanding Logarithmic expressions

(c) $\ln \dfrac{3x^2}{(x+1)^{10}}$

Expand the expression:

$\ln \frac{3x^2}{(x+1)^{10}}$

Apply the Second Law of Logarithms

$\ln \frac{3x^2}{(x+1)^{10}} = \ln 3x^2 - \ln(x+1)^{10}$

Apply the First Law of Logarithms for $\ln 3x^2$

$\ln(3 \times x^2) = \ln 3 + \ln x^2$

$\ln 3 + \ln x^2 - \ln(x+1)^{10}$

Apply the Third Law of Logarithms for $\ln x^2$ and $\ln(x+1)^{10}$

$\ln x^2 = 2\ln x$
$\ln(x+1)^{10} = 10\ln(x+1)$

Assemble the expression
$\ln 3 + 2\ln x - 10\ln(x+1)$

SOLUTION

Combining Logarithmic expressions

(a) $\frac{1}{2}\log_2 5 - 2\log_2 7$

(b) $\log_5(x^2 - 1) - \log_5(x - 1)$

ACTIVITY 1.3

1.3 Combining Logarithmic expressions

(a) $\frac{1}{2}\log_2 5 - 2\log_2 7$

Combine the expression:
$\frac{1}{2}\log_2 5$ - $2\log_2 7$

Apply the Third Law of Logarithms for $\frac{1}{2}\log_2 5$ and $2\log_2 7$

$\frac{1}{2}\log_2 5 = \log_2 5^{\frac{1}{2}}$
$2\log_2 7 = \log_2 7^2$

$\log_2 5^{\frac{1}{2}}$ - $\log_2 7^2$
$\log_2 \sqrt{5}$ - $\log_2 7^2$ [Note: $5^{\frac{1}{2}} = \sqrt{5}$]

Apply the Second Law of Logarithms
$\log_2 \sqrt{5}$ - $\log_2 7^2 = \log_2\left(\frac{\sqrt{5}}{7^2}\right)$

$\log_2\left(\frac{\sqrt{5}}{49}\right)$

(b) $\log_5(x^2 - 1) - \log_5(x - 1)$

Combine the expression:
$\log_5(x^2-1)$ - $\log_5(x-1)$

Apply the Second Law of Logarithms

$\log_5(x^2-1)$ - $\log_5(x-1) = \log_5 \frac{(x^2-1)}{(x-1)}$

We know that x^2-1 is a difference of two squares since x^2 and 1 are perfect squares.

Factor (x^2-1)

$\log_5 \frac{(x-1)(x+1)}{(x-1)}$

Simplify

$\log_5(x+1)$

SOLUTION

2 Exponential and Logarithmic equations

Exponential equations

An ***exponential equation*** is one in which the **variable occurs in the exponent**.

Some exponential equations can be solved by using the fact that exponential functions are **one-to-one**.

$$a^x = a^y \quad \Rightarrow \quad x = y$$

EXAMPLE

Solve the exponential equation.

(a) $5^x = 125$ **(b)** $5^{2x} = 5^{x+1}$

SOLUTION

(a) We first express 125 as a power of 5 and then use the fact that the exponential function $f(x) = 5^x$ is one-to-one.

$5^x = 125$	Given equation
$5^x = 5^3$	Because $125 = 5^3$
$x = 3$	One-to-one property

The solution is $x = 3$.

(b) We first use the fact that the function $f(x) = 5^x$ is one-to-one.

$5^{2x} = 5^{x+1}$	Given equation
$2x = x + 1$	One-to-one property
$x = 1$	Solve for x

The solution is $x = 1$.

We solved the equation here by **comparing the exponents** on both sides.

Not suitable for more **complex problems**!

EXAMPLE

Guideline for solving exponential equations

GUIDELINES FOR SOLVING EXPONENTIAL EQUATIONS

1. Isolate the exponential expression on one side of the equation.
2. Take the logarithm of each side, then use the Laws of Logarithms to "bring down the exponent."
3. Solve for the variable.

Solving an exponential equation

Consider the exponential equation $3^{x+2} = 7$.

(a) Find the exact solution of the equation expressed in terms of logarithms.

We take the common logarithm of each side and use Law 3.

$$
\begin{aligned}
3^{x+2} &= 7 && \text{Given equation} \\
\log(3^{x+2}) &= \log 7 && \text{Take log of each side} \\
(x+2)\log 3 &= \log 7 && \text{Law 3 (bring down exponent)} \\
x + 2 &= \frac{\log 7}{\log 3} && \text{Divide by log 3} \\
x &= \frac{\log 7}{\log 3} - 2 && \text{Subtract 2}
\end{aligned}
$$

The exact solution is $x = \dfrac{\log 7}{\log 3} - 2$.

EXAMPLE

Solving an exponential equation

Solve the equation $8e^{2x} = 20$.

SOLUTION We first divide by 8 to isolate the exponential term on one side of the equation.

$$
\begin{aligned}
8e^{2x} &= 20 && \text{Given equation} \\
e^{2x} &= \tfrac{20}{8} && \text{Divide by 8} \\
\ln e^{2x} &= \ln 2.5 && \text{Take ln of each side} \\
2x &= \ln 2.5 && \text{Property of ln} \\
x &= \frac{\ln 2.5}{2} && \text{Divide by 2 (exact solution)} \\
&\approx 0.458 && \text{Calculator (approximate solution)}
\end{aligned}
$$

EXAMPLE

Exponential equation of quadratic type

Solve the equation $e^{2x} - e^x - 6 = 0$.

SOLUTION To isolate the exponential term, we factor.

$$\begin{aligned} e^{2x} - e^x - 6 &= 0 && \text{Given equation} \\ (e^x)^2 - e^x - 6 &= 0 && \text{Law of Exponents} \\ (e^x - 3)(e^x + 2) &= 0 && \text{Factor (a quadratic in } e^x\text{)} \end{aligned}$$

$$e^x - 3 = 0 \quad \text{or} \quad e^x + 2 = 0 \qquad \text{Zero-Product Property}$$

$$e^x = 3 \qquad\qquad e^x = -2$$

The equation $e^x = 3$ leads to $x = \ln 3$. But the equation $e^x = -2$ has no solution because $e^x > 0$ for all x. Thus $x = \ln 3 \approx 1.0986$ is the only solution. You should check that this answer satisfies the original equation.

EXAMPLE

An Equation Involving Exponential Functions

Solve the equation $3xe^x + x^2e^x = 0$.

SOLUTION First we factor the left side of the equation.

$$3xe^x + x^2e^x = 0 \qquad \text{Given equation}$$

$$x(3 + x)e^x = 0 \qquad \text{Factor out common factors}$$

$$x(3 + x) = 0 \qquad \text{Divide by } e^x \text{ (because } e^x \neq 0\text{)}$$

$$x = 0 \qquad \text{or} \qquad 3 + x = 0 \qquad \text{Zero-Product Property}$$

Thus the solutions are $x = 0$ and $x = -3$.

EXAMPLE

Exponential equations

(a) $10^{2x-3} = \frac{1}{10}$

(b) $10^{2x^2-3} = 10^{9-x^2}$

ACTIVITY 2.1

2.1 Exponential equations

(a) $10^{2x-3} = \frac{1}{10}$

$10^{2x-3} = \dfrac{1}{10}$

Rewrite this equation as $10^{2x-3} = 10^{-1}$:

$10^{2x-3} = 10^{-1}$

Use the one-to-one property to solve this equation:

$2x - 3 = -1$

Solve for x:

$2x = -1 + 3$

$2x = 2$

$x = \dfrac{2}{2}$

$x = 1$

(b) $10^{2x^2-3} = 10^{9-x^2}$

$10^{2x^2-3} = 10^{9-x^2}$

Simply use the one-to-one property to solve this equation. Make the exponents on both sides of the equation equal:

$2x^2 - 3 = 9 - x^2$

Take the $-x^2$ to the left side of the equation and take -3 to the right side:

$2x^2 + x^2 = 9 + 3$

Simplify both sides:

$3x^2 = 12$

Take the 3 to divide the right side:

$x^2 = \dfrac{12}{3}$

$x^2 = 4$

Take the square root of both sides of the equation:

$\sqrt{x^2} = \sqrt{4}$

$x = \pm 2$

SOLUTION

Solve in terms of logarithms

(a) $2e^{12x} = 17$

(b) $1 + e^{4x+1} = 20$

(c) $e^{2x} - e^{x} - 6 = 0$

ACTIVITY 2.2

2.2 a,b: Solve in terms of logarithms

(a) $2e^{12x} = 17$

$2e^{12x} = 17$

Take the 2 to divide the right side of the equation:

$e^{12x} = \frac{17}{2}$

Apply $\ln$ to both sides:

$\ln e^{12x} = \ln\frac{17}{2}$

Take down the exponent $12x$ to multiply in front of the $\ln$:

$12x \ln e = \ln\frac{17}{2}$

Since $\ln e = 1$, the equation becomes:

$12x = \ln\frac{17}{2}$

$$x = \frac{1}{12}\ln\left(\frac{17}{2}\right)$$

(b) $1 + e^{4x+1} = 20$

$1 + e^{4x+1} = 20$

First, let's solve for e^{4x+1}:

$e^{4x+1} = 20 - 1$

$e^{4x+1} = 19$

Apply $\ln$ to both side of the equation:

$\ln e^{4x+1} = \ln 19$

The exponent $4x+1$ can be taken down to multiply in front of its respective $\ln$:

$(4x+1)\ln e = \ln 19$

Since $\ln e = 1$, the equation becomes:

$4x + 1 = \ln 19$

Solve for x:

$4x = \ln 19 - 1$

$$x = \frac{1}{4}(\ln 19 - 1)$$

SOLUTION

2.2 c: Solve in terms of logarithms

(c) $e^{2x} - e^{x} - 6 = 0$

$e^{2x} - e^{x} - 6 = 0$
Let $e^{x} = u$ and $e^{2x} = u^2$ and rewrite the equation:
$u^2 - u - 6 = 0$
Solve by factoring:
$(u-3)(u+2) = 0$
We get two solutions, which are:
$u = 3$ and $u = -2$
Let's undo the initial substitution, knowing that
$u = e^{x}$. The solutions then become:
$e^{x} = 3$ and $e^{x} = -2$
Solve for x:
$x = \ln 3 \approx 1.098612$
The other equation has no real solution.

SOLUTION

Logarithmic equations

A *logarithmic equation* is one in which a logarithm of the variable occurs.

GUIDELINES FOR SOLVING LOGARITHMIC EQUATIONS

1. Isolate the logarithmic term on one side of the equation; you might first need to combine the logarithmic terms.
2. Write the equation in exponential form (or raise the base to each side of the equation).
3. Solve for the variable.

Solving logarithmic equations

Solve each equation for x.

(a) $\ln x = 8$

(b) $\log_2(25 - x) = 3$

SOLUTION

(a)

$$\ln x = 8 \qquad \text{Given equation}$$

$$x = e^8 \qquad \text{Exponential form}$$

Therefore $x = e^8 \approx 2981$.

We can also solve this problem another way.

$$\ln x = 8 \qquad \text{Given equation}$$

$$e^{\ln x} = e^8 \qquad \text{Raise } e \text{ to each side}$$

$$x = e^8 \qquad \text{Property of ln}$$

(b) The first step is to rewrite the equation in exponential form.

$$\log_2(25 - x) = 3 \qquad \text{Given equation}$$

$$25 - x = 2^3 \qquad \text{Exponential form (or raise 2 to each side)}$$

$$25 - x = 8$$

$$x = 25 - 8 = 17$$

EXAMPLE

Solving logarithmic equations

Solve the equation $4 + 3\log(2x) = 16$.

SOLUTION We first isolate the logarithmic term. This allows us to write the equation in exponential form.

$4 + 3\log(2x) = 16$	Given equation
$3\log(2x) = 12$	Subtract 4
$\log(2x) = 4$	Divide by 3
$2x = 10^4$	Exponential form (or raise 10 to each side)
$x = 5000$	Divide by 2

EXAMPLE

Solving logarithmic equations

Solve the equation $\log(x+2)+\log(x-1)=1$ algebraically and graphically.

CHECK YOUR ANSWERS

$x=-4$:

$$\log(-4+2)+\log(-4-1)$$
$$=\log(-2)+\log(-5)$$
$$\text{undefined} \quad ✗$$

$x=3$:

$$\log(3+2)+\log(3-1)$$
$$=\log 5+\log 2=\log(5\cdot 2)$$
$$=\log 10=1 \quad ✓$$

SOLUTION 1: Algebraic

We first combine the logarithmic terms, using the Laws of Logarithms.

$$\begin{aligned} \log[(x+2)(x-1)] &= 1 && \text{Law 1} \\ (x+2)(x-1) &= 10 && \text{Exponential form (or raise 10 to each side)} \\ x^2+x-2 &= 10 && \text{Expand left side} \\ x^2+x-12 &= 0 && \text{Subtract 10} \\ (x+4)(x-3) &= 0 && \text{Factor} \\ x=-4 \quad \text{or} \quad x &= 3 \end{aligned}$$

We check these potential solutions in the original equation and find that $x=-4$ is not a solution (because logarithms of negative numbers are undefined), but $x=3$ is a solution. (See *Check Your Answers*.)

EXAMPLE

Solve in terms of logarithms

(a) $\log_5 x + \log_5(x + 1) = \log_5 20$

(b) $\ln\left(x - \frac{1}{2}\right) + \ln 2 = 2 \ln x$

(c) $\log_2(x^2 - x - 2) = 2$

ACTIVITY 2.3

2.3: Solve in terms of logarithms

(a) $\log_5 x + \log_5(x+1) = \log_5 20$

Combine the logarithms on the left side:

$\log_5 x(x+1) = \log_5 20$
$\log_5(x^2+x) = \log_5 20$

Since log is one to one, this equation becomes:

$x^2 + x = 20$

Subtract 20 from each side:

$x^2 + x - 20 = 0$

Solve by factoring:

$(x-4)(x+5) = 0$

We get two solutions, which are:

$x = 4$ and $x = -5$

If we check our answers, we'll see that $x = -5$ is not a solution to this equation because $\log_5(-5)$ is not defined.

.

So our final answer is $x = 4$.

(b) $\ln\left(x - \frac{1}{2}\right) + \ln 2 = 2\ln x$

Combine the logarithms on the left as a product and rewrite the expression on the right as a power:

$\ln 2\left(x - \frac{1}{2}\right) = \ln x^2$

$\ln(2x-1) = \ln x^2$

Since $\ln$ is one to one, the equation becomes:

$2x - 1 = x^2$

Take all terms to the right side:

$x^2 - 2x + 1 = 0$

Solve by factoring:

$(x-1)^2 = 0$

We get one solution:

$x = 1$

SOLUTION

2.3: Solve in terms of logarithms

(c) $\log_2(x^2 - x - 2) = 2$

Write this equation in exponential form:

$x^2 - x - 2 = 2^2$

$x^2 - x - 2 = 4$

Take the 4 to the left side of the equation:

$x^2 - x - 2 - 4 = 0$

$x^2 - x - 6 = 0$

Solve by factoring:

$(x - 3)(x + 2) = 0$

We get two solutions:

$x = 3$ and $x = -2$

SOLUTION

Word problems

(I) **Fish Population** A small lake is stocked with a certain species of fish. The fish population is modeled by the function

$$P = \frac{10}{1 + 4e^{-0.8t}}$$

where P is the number of fish in thousands and t is measured in years since the lake was stocked.

(a) Find the fish population after 3 years.

(b) After how many years will the fish population reach 5000 fish?

(II) **Atmospheric Pressure** Atmospheric pressure P (in kilopascals, kPa) at altitude h (in kilometers, km) is governed by the formula

$$\ln\left(\frac{P}{P_0}\right) = -\frac{h}{k}$$

where $k = 7$ and $P_0 = 100$ kPa are constants.

(a) Solve the equation for P.

(b) Use part (a) to find the pressure P at an altitude of 4 km.

ACTIVITY 2.4

2.4: Solve in terms of logarithms

(I) **(a)** $P(3) = \dfrac{10}{1 + 4e^{-0.8(3)}} = 7.337$, so there are approximately 7337 fish after 3 years.

(b) We solve for t. $\dfrac{10}{1 + 4e^{-0.8t}} = 5 \Leftrightarrow 1 + 4e^{-0.8t} = \frac{10}{5} = 2 \Leftrightarrow 4e^{-0.8t} = 1 \Leftrightarrow e^{-0.8t} = 0.25 \Leftrightarrow -0.8t = \ln 0.25 \Leftrightarrow$

$t = \dfrac{\ln 0.25}{-0.8} = 1.73$. So the population will reach 5000 fish in about 1 year and 9 months.

(II) **(a)** $\ln\left(\dfrac{P}{P_0}\right) = -\dfrac{h}{k} \Leftrightarrow \dfrac{P}{P_0} = e^{-h/k} \Leftrightarrow P = P_0 e^{-h/k}$. Substituting $k = 7$ and $P_0 = 100$ we get $P = 100e^{-h/7}$.

(b) When $h = 4$ we have $P = 100e^{-4/7} \approx 56.47$ kPa.

SOLUTION